

Table 10.2: Performance Benchmarks

Metric	Target	Measured	Status
Initial Page Load (Landing)	< 3 seconds	1.4 seconds	Pass
Dashboard Load (Farmer)	< 3 seconds	1.8 seconds	Pass
Dashboard Load (Admin)	< 3 seconds	2.1 seconds	Pass
Firestore Query Latency	< 500ms	180ms (avg)	Pass
Image Upload (Cloudinary)	< 5 seconds	2.3 seconds (avg)	Pass
Razorpay Checkout Modal	< 2 seconds	1.1 seconds	Pass
Carbon Credit Calculation	< 100ms	5ms	Pass
TypeScript Compilation	< 30 seconds	12 seconds	Pass

10.4 Testing Results Summary

All 30 test cases passed successfully. Key findings from the testing process:

- **Zero Critical Bugs:** No critical or blocking issues were found during the final testing phase.
- **Type Safety:** TypeScript's strict mode caught 23 potential runtime errors during compilation that would have been undetectable in plain JavaScript.
- **Security Compliance:** All Firestore security rules properly enforce role-based access. No unauthorized data access was possible during penetration testing.
- **Performance:** All pages load within the 3-second target on standard broadband connections. Firestore query performance averages 180ms across all tested operations.
- **Cross-Browser:** The application renders consistently across all tested browsers with no layout breaking or functionality loss.